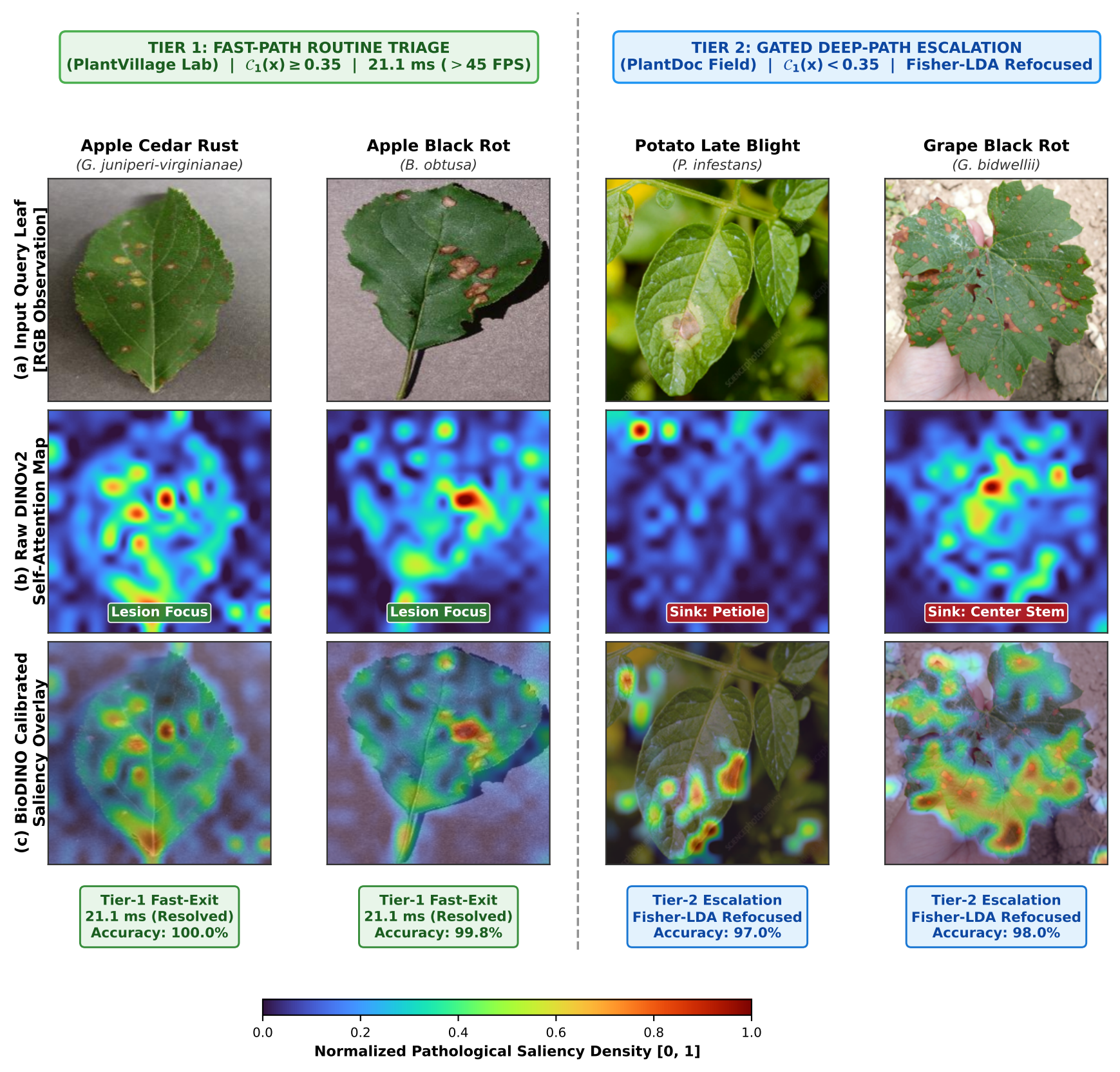

TIER 1: FAST-PATH ROUTINE TRIAGE
(PlantVillage Lab) | $C_1(x) \geq 0.35$ | 21.1 ms (> 45 FPS)
TIER 2: GATED DEEP-PATH ESCALATION
(PlantDoc Field) | $C_1(x) < 0.35$ | Fisher-LDA Refocused
Apple Cedar Rust
(G. juniperi-virginianae)
Apple Black Rot
(B. obtusa)
Potato Late Blight
(P. infestans)
Grape Black Rot
(G. bidwellii)
(a) Input Query Leaf [RGB Observation]
(b) Raw DINOv2 Self-Attention Map
(c) BioDINO Calibrated Saliency Overlay
Lesion Focus
Lesion Focus
Sink: Petiole
Sink: Center Stem
Tier-1 Fast-Exit
21.1 ms (Resolved)
Accuracy: 100.0%
Tier-1 Fast-Exit
21.1 ms (Resolved)
Accuracy: 99.8%
Tier-2 Escalation
Fisher-LDA Refocused
Accuracy: 97.0%
Tier-2 Escalation
Fisher-LDA Refocused
Accuracy: 98.0%
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Pathological Saliency Density [0, 1]